**Department of Chemical Engineering**

**Chemical Reaction Engineering II (CL 24303)**

MO2026

Tutorial 0

| Date Assigned | 30.07.2026 | Submission Date and Time | 09.08.2026 23:59 |
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| 1. | Define the rate of reaction. | 1 |
| 2. | What is Arrhenius' Law? Explain with the required expression | 2 |
| 3. | If $-r_A = -\left(\frac{dC_A}{dt}\right) = 0.2$ mol/lit-sec, when $C_A = 1$ mol/lit, what is the rate of reaction when $C_A = 10$ mol/lit if the order of the reaction is not known? | 1 |
| 4. | What is the difference between space-time and residence time? | 2 |
| 5. | Aqueous A reacts to form R ($A \rightarrow R$) and in the first minute in a batch reactor, its concentration drops from $C_{Ao}$ = 2.03 mol/liter to $C_{Af}$ = 1.97 mol/liter. Find the rate equation for the reaction if the kinetics are second order with respect to A. | 2 |
| 6. | One liter per minute of liquid containing A and B ($C_{A0}$= 0.10 mol/liter, $C_{B0}$ =0.01 mol/liter) flow into a mixed reactor of volume V = 1 liter. The materials react in a complex manner for which the stoichiometry is unknown. The outlet stream from the reactor contains A, B, and C ($C_{Af}$ = 0.02 mol/liter, $C_{Bf}$ = 0.03 mol/liter, $C_{cf}$ = 0.04 mol/liter). Find the rate of reaction of A, B, and C for the conditions within the reactor. | 2 |
| 7. | Consider a reaction being carried out in a real reactor that can be modelled as two different reactor systems. To simplify the calculations, let $\tau_{PFR} = \tau_{CSTR}$ = 1 min, let the reaction rate constant equal 1.0 $m^3$/kmol min, and let the initial concentration of liquid reactant, $C_{A0}$, equal 1.0 kmol/$m^3$. Find the conversion for the following system when the reaction order is n=1 and 2<br>a) In the first system, an ideal CSTR is followed by an ideal PFR<br>b) In the second system, an ideal PFR is followed by an ideal CSTR<br>Comment on the result | 5 |